

MODEL PARAMETERS AND ITS SIZE FOR TRAINING AND MODEL FILE

For Calculation:	NAME	DIMS			TOTAL VALUES	SIZE	UNIT
ATTENTION	<i>KdotQ</i>	4096	4096	1	16777216	64	MB
MATRIX	<i>Key (K)</i>	4096	64	1	262144	1	
	<i>Query (Q)</i>	4096	64	1	262144	1	
	<i>Horizontal (H)</i>	64	4096	1	262144	1	
	<i>Vertical (V)</i>	64	4096	1	262144	1	
MLP	<i>Hor</i>	64	64	16	65536	0.25	
	<i>Ver</i>	64	64	16	65536	0.25	
Continuous Calculation for every head	<i>KEYS</i>	4096	64	1	262144	1	
	<i>QUERIES</i>	4096	64	1	262144	1	
	<i>EV</i>	4096	64	1	262144	1	
	<i>Hlayers</i>	64	16	2	2048	0.0078125	
	<i>Activations</i>	64	16	2	2048	0.0078125	
	<i>gweights (hor)</i>	64	64	16	65536	0.25	
	<i>gweights (ver)</i>	64	64	16	65536	0.25	
Complete Model Training Parameters and Size (one parallel is number of attention heads working in parallel)				Attention	18878464	72.015625	
				One Parallel:	151027712	576.125	
				Model:	38663094272	144.03125	GB
Very Small	<i>TOKENS</i>	1	64	32768	2097152	8	MB
	<i>1D</i>	3	64	2048	393216	1.5	
				TOTAL:	38665584640	144.0405273	GB

Data type	float
Bytes	4
Embedding	64
Layers (PAs)	8
Blocks (CAs)	8
Attention Per layer (IAs)	32
Layers of MLP	16
Context	4096
Height	4096
Number of MLP/Head	2
Incomplete Attention IA	Single Head
Partial Attention (Layer)	Consecutive IAs
Block (Complete Attention)	Parallel PAs
Transformer	Multi or Single CA
One Parallel	Heads in parallel ops.

For Calculation:	NAME	DIMS			TOTAL VALUES	SIZE	UNIT
ATTENTION	<i>KdotQ</i>	4096	4096	1	16777216	64	MB
CACHE	<i>QK'</i>	64	64	1	4096	0.015625	
	<i>QV'</i>	64	64	1	4096	0.015625	
	<i>KH'</i>	64	64	1	4096	0.015625	
MLP	<i>Hor</i>	64	64	16	65536	0.25	
	<i>Ver</i>	64	64	16	65536	0.25	
Continuous Calculation for every head	<i>KEYS</i>	4096	64	1	262144	1	
	<i>QUERIES</i>	4096	64	1	262144	1	
	<i>Hlayers</i>	64	16	2	2048	0.0078125	
	<i>Activations</i>	64	16	2	2048	0.0078125	
	<i>gweights (hor)</i>	64	64	16	65536	0.25	
	<i>gweights (ver)</i>	64	64	16	65536	0.25	
For Model Use Parameters and Size				Attention:	17580032	67.0625	
				One Parallel:	140640256	536.5	
				Model:	4500488192	16.765625	GB
Very Small	<i>TOKENS</i>	1	64	32768	2097152	8	MB
	<i>EV</i>	1	64	8388608	536870912	2048	
	<i>1D</i>	3	64	256	49152	0.1875	
				TOTAL:	5039505408	18.77362061	GB

NAME	PARAMETERS	SIZE (MB)	GB
MLP	268435456	1024	1
MAT	2147483648	8192	8
CACHE	25165824	96	0.09375
FOR TRAINING	2415919104	9216	9
FOR USE	293601280	1120	1.09375
TOTAL	2441084928	9312	9.09375

For Trained Params:	NAME	DIMS			Attention	Model	SIZE	UNIT
MATRIX	<i>Key (K)</i>	4096	64	1	262144	536870912	2048	MB
	<i>Query (Q)</i>	4096	64	1	262144	536870912	2048	
	<i>Horizontal (H)</i>	64	4096	1	262144	536870912	2048	
	<i>Vertical (V)</i>	64	4096	1	262144	536870912	2048	
MLP	<i>Hor</i>	64	64	16	65536	134217728	512	
	<i>Ver</i>	64	64	16	65536	134217728	512	
CACHE	<i>QK'</i>	64	64	1	4096	8388608	32	
	<i>QV'</i>	64	64	1	4096	8388608	32	
	<i>KH'</i>	64	64	1	4096	8388608	32	
When Model is being trained: Parameters and Size (embeddings excluded)					1191936	2441084928	9312	